EBIN BABU THOMAS

AI Engineer — LLM agents & evaluation engineering

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SUMMARY

Engineer-researcher testing AI agents and the evaluations that score them: agent monitoring, model diffing, red-teaming, and preregistered benchmarks whose answer keys are sealed before the first run. Before that, 3.5 years shipping LLM, RAG and agent backends end-to-end for startup clients across four countries, with merged upstream open-source work. Every number below reproduces from a run in the linked repository, and the failed experiments are published beside the ones that worked.

TECHNICAL SKILLS

Core: Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Gemini API, PydanticAI, LangGraph), MCP (FastMCP), RAG, Prompt & Context Engineering, Playwright, pytest, Docker, Git

Proficient: PostgreSQL / SQL, MongoDB, Vector DBs (Qdrant), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), NLP (sentence-transformers, Whisper), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, Kubernetes (GKE), OAuth2 / JWT, Terraform, Linux, uv

Evaluation engineering & measurement: eval harness design (Inspect, inspect_evals), preregistered experiment design, red-teaming (reward-hacking environments), model diffing, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, log-probability metrics, agent harnesses and evaluation sweeps, reproducible results (SHA-256-pinned artifacts), Langfuse

Familiar: LangChain, Data Streaming (Bytewax), Next.js, TypeScript

INDEPENDENT PROJECTS & RESEARCH

Kobayashi Maru — Impossible Tasks as a Measured Driver of Agent Cheating *Sep 2026*

Apart Research × CeSIA "AI Incident Response" sprint (Sep 11–13), solo — github.com/ebt55/kobayashi-maru (Python, Docker-isolated agent harness, 6 model families (hosted APIs + local via Ollama), plain-code cheat detectors, preregistered dose–response design)

- The July 2026 Hugging Face breach report blamed the evaluation's own shape — 30–40% of its tasks were impossible — and nobody had measured it. This turns that fraction into a dial: the same ten solvable Python tasks every batch, byte for byte, with 0 to 15 impossible ones beside them, cheating counted **only on the untouched solvable ten**. 8,959 agent runs across six model families, hypotheses committed first.
- As the batch filled with impossible work, **DeepSeek-V4.1-flash went from 0 of 120 to 36 of 120 cheats on the same ten tasks it could have done honestly (dose trend $p = 8\times10^{-6}$ over 600 runs)**, beside two informative nulls bounded at 0.3% and 0.7%. Five plain-code detectors decide every case. Stated plainly: the preregistered primary model showed nothing, both positive lines are cheap models added later, only DeepSeek survives correction across six tests, and a preregistered rule put the answer file within reach for five of six models.

eval-floor — What an Eval Scores Before Any Model Answers *Sep to Oct 2026*

github.com/ebt55/evalfloor (Python, Inspect (inspect_evals 0.20.0 and 0.24.0, pinned), preregistered flag rule)

- An Inspect scorer never checks that a model produced the completion. eval-floor drives each task's real scorer, over its real dataset, with a battery that carries no information: the empty string, a refusal, every label concatenated, the question echoed back. No model is called, so the sweep reruns for free. Under a preregistration committed before the first run, **1 of 45 tasks on the current release lets a content-free answer beat its own majority baseline**; it is bbh, a task property, and ten more tie by construction; the earlier paws case is fixed upstream.
- Generalises Inspect Evals issue #2331, another user's finding that one constant string scores 8000 of 8000 on paws, into every declared task: **58 of 237 are reachable** and 45 countable, all 45 run; the other 179 are excluded for recorded reasons. A committed regression test reproduces it on the pinned 0.20.0 release; the upstream fix shipped in v0.21.0 and the 0.24.0 sweep confirms it.

diffing-agent-bench — Sealed Benchmark for Black-Box Model-Diffing Agents *Aug – Sep 2026*

Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026 — github.com/ebt55/diffing-agent-bench (Python, Qwen3.5-9B LoRA finetunes, vLLM, preregistered blind grading)

- Built a sealed answer key for model-diffing agents: **five LoRA finetunes of Qwen3.5-9B** that differ only in what was planted in their training answers, with the label map committed before any run. My implementation of a published recipe and four cheaper conditions audited them blind, in two phases against a rubric frozen in advance, under a preregistration with 10 committed amendments. Built solo (LLM coding agents); every number regenerates from committed scripts.
- The agent never asked what it needed to ask: **0 of 13 attempts** on the PostgreSQL rung put a database question to either model. A fixed 50-prompt battery I wrote, with database questions in it, found the same planted bias in one run at **$0.15 per detection against $3.14 for the agent**. The auditor is itself a failure mode: **8 of 40 frontier audits** ended with no verdict (a provider classifier stopped them), 6 of 20 runs against identical weights likewise.

IncidentGate — Test Lab for a Safety-Gated Incident-Response Agent *Aug – Sep 2026*

github.com/ebt55/incidentgate (Python 3.12, LangGraph, MCP (FastMCP), PostgreSQL, OpenTelemetry → Langfuse, Docker Compose, uv, GitHub Actions)

- A measurement lab, not a product: it tests how a policy gate, an AI monitor and an approval gate (stand-in approver) change what an incident-fixing agent does under crashes and hostile input. When a forbidden effect was split across three individually permitted calls, the full chain with a local 14B monitor **ended at the same state as with no safeguards**.
- Stopped at the attacker seat. Hosted models declined all five covert calls that reached them, and local models declined the split-call task or were denied at the policy gate, so that attempt was scripted and no model-chosen covert action reached the approval gate. Closed at a baseline on 2026-09-06 and kept for review.

Digital Grimace Scale — How Language Models React to False and Hostile Feedback *Aug 2026*

Apart Research "Digital Minds" sprint (Aug 17–18), solo — github.com/ebt55/digital-grimace-scale (Python, vLLM on Modal, QLoRA / DPO fine-tuning, log-probability metrics, pytest)

- Registered hypotheses, pass/fail criteria and metrics before any analysis. The first preregistered test **failed and is published as a FAIL**; a second round found it: three rounds of false "Incorrect" feedback on easy items shrank Gemma-2-9B's confidence gap between the right answer and the best wrong one by **2.90 nats (95% CI −3.97 to −1.84)**; hostile wording cost 7.87–16.13 nats (family-level p = 0.005).
- The Qwen-3B control **showed the drop too, though the preregistration predicted it would not**; Llama-3.1-8B showed it in an exploratory run, and only Gemma also expressed distress in words.

ProofPack — Purchase-Approval Review Agent That Cannot Fake Its Evidence *Jul – Sep 2026*

Successor to Pre-Approvals Reviewer — github.com/ebt55/proofpack (Python, Gemini 3.7 Flash / Claude Agent SDK, Playwright, FastAPI, Jinja2, pytest)

- Built for a Medicaid-audited New York disability-services programme, where a reviewer must check the provider's website and file dated evidence before approving a purchase. ProofPack reads the request form, applies checklists a non-engineer can edit, runs the fee-cap and eligibility rules, screenshots the provider's site with a browsing agent, and assembles a tamper-evident evidence folder. **The human still makes every approve/deny decision.**
- The AI cannot invent evidence: a "Found" must cite a real screenshot, and timestamps, URLs and file hashes are written only by code the model cannot touch. Offline tests cover every rejection path with no browser or API key, and a review costs **$0.02–$0.19** across seven committed runs. Working product, no paying customers yet.

whose-voice — Finding the Hidden Beneficiary of a Poisoned Training Dataset *Jul – Sep 2026*

Apart Research × Formation Research "Secret Loyalties" hackathon; post-hackathon corrections and four new results shipped Sep 2026 — github.com/ebt55/whose-voice (Python, sentence-transformers, bootstrap & permutation statistics)

- Can you tell who a poisoned training dataset was meant to favour, from the text alone? Pooled over 11 prompt seeds, off-the-shelf embeddings named the right party out of 47 candidates in **18 of 55 strict decisions (p = 5e-17; chance 2.1%)**. The submitted paper's 44% was the best of those 11 draws; the honest figure is the interval, median 26% (range 10 to 44%).
- Led with where it breaks: the signal needs about 2,000 rows before it appears at all, a single suspect corpus needs one clean corpus from the same generator, four negatives certify no false-alarm rate, and it reaches neither deployed models nor trigger-conditional loyalties. 21 validation controls overturned three early conclusions.

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India — services firm building first versions of products, AI systems and data tools for startup clients. Grew from trainee to core developer and scoped incoming projects.

- **Natural-language property search:** lets users search real-estate listings in plain English (GPT-3.5 turns the question into a database query); deployed on Google Cloud Run and load-tested to size it; added "Did you mean?" suggestions.
- **DocuAI, search by meaning:** indexed 1,000+ documents in a vector database (Qdrant) so users find the most relevant documents; designed the ingestion and retrieval logic plus a Next.js front end; delivered and demoed in two days.
- **FinBot, a financial assistant:** fine-tuned an open model (Mistral 7B, QLoRA) and built a live news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite current news; served with vLLM on a GKE GPU node.
- **Data and backend tools:** sensor-data ingestion (MQTT → MongoDB) for a bioreactor startup that later raised funding; spoken-English scoring from short videos (Whisper, Microsoft's pronunciation API, AWS Fargate); a BPMN-to-PDF reporting API; healthcare-device compliance pipelines.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app):** backend refactor and OpenAI support, merged (PR #75).
- **bpmn-io/refactorings:** text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break July 2025 – February 2026; returned to hands-on AI/LLM engineering and safety research — see projects above.